Multiple testing with e-processes

5th AI & Mathematics workshop, 4 June 2026

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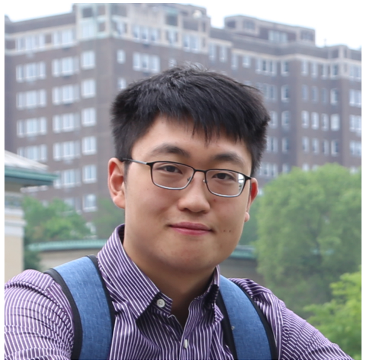
Joint work with...



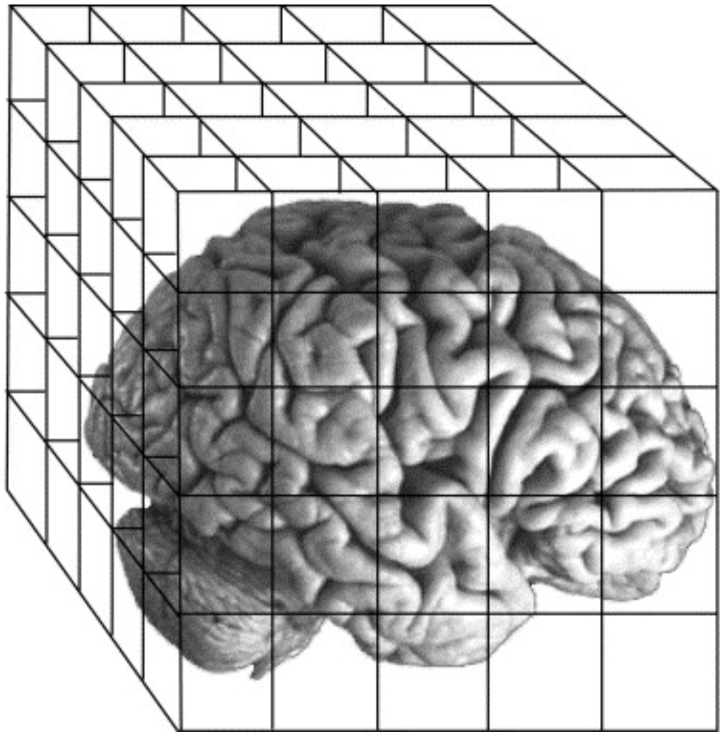


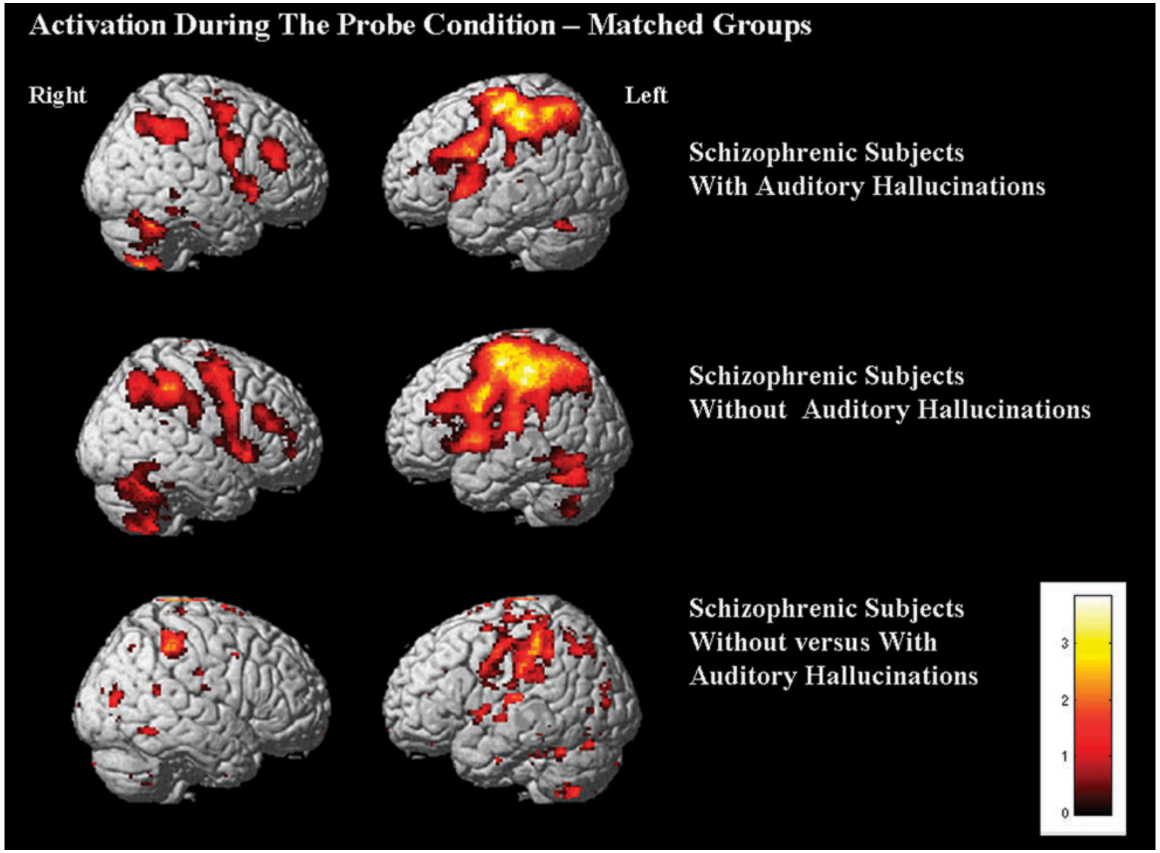




Example: multiple testing in neuroimaging

130.000 voxels





Wible, Cynthia G. et al. "fMRI activity correlated with auditory hallucinations during performance of a working memory task: data from the FBIRN consortium study." Schizophrenia bulletin 35 1 (2009): 47-57 .

Bringing flexibility to multiple testing

- Researchers want to work interactively with the data, which is not possible with current methods

- How can this be achieved? New theory of hypothesis testing with e-values and e-processes

E-values and e-processes

The e-variable

- **Definition**: e-variable

  An e-variable E for $\mathscr{P}$ is a non-negative random variable satisfying $\mathbb{E}_P[E] \leq 1$ for all $P \in \mathscr{P}$.

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  An e-variable E for $\mathscr{P}$ is a non-negative random variable satisfying $\mathbb{E}_P[E] \leq 1$ for all $P \in \mathscr{P}$.

- The value taken by the e-variable after observing the data is called the e-value. However, often, as also happens with the infamous p-value (p-variable), the random variable E itself is also often called e-value.

Tests and the type I error guarantee

- **Definition:** binary test

  A binary test ϕ is a $\{0,1\}$-valued random variable. The type-I error of a test ϕ for P is $\mathbb{E}_P[\phi]$. A test has level $\alpha \in [0,1]$ for $\mathscr{P}$ if its type-I error is at most α for every $P \in \mathscr{P}$.

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- **Markov's inequality for e-variables**

  Let E be an e-variable for $\mathscr{P}$. We have $P(E \geq 1/\alpha) \leq \alpha$ for all $P \in \mathscr{P}$ and $\alpha \in (0,1]$. Hence, $\mathbf{1}_{\{E \geq 1/\alpha\}}$ is a binary test of level α.

The e-process: any-time valid inference (1)

- Any-time valid inference is a new paradigm for sequential testing (or estimation) that allows the statistician to stop at any arbitrary stopping time, possibly not anticipated or specified in advance.

- E-processes are the central object for any-time valid inference

The e-process: any-time valid inference (2)

- E-processes are the central object for any-time valid inference
- Filtrations are important
- A filtration is an increasing nested sequence of sigma-algebra's, which we denote with $\mathfrak{F} := (\mathcal{F}_t)_{t \geq 0}$.
- A sequence of random variables $(Y_t)_{t \geq 0}$ is called a process if it is adapted to a filtration $\mathfrak{F}$, that means that every Y_t is measurable with respect to $\mathcal{F}_t$.
- A stopping time (or rule) τ is a nonnegative integer valued random variable such that $\{\tau \leq t\} \in \mathcal{F}_t$ for each $t \geq 0$. Let $\mathcal{T}$ be the set of all stopping times.

The e-process: any-time valid inference (3)

- **Definition:** e-process

  A nonnegative process E is called an e-process if it satisfies $\mathbb{E}_P[E_\tau] \leq 1$ for any stopping time $\tau \in \mathscr{T}$ and any $P \in \mathscr{P}$.

- (There is an equivalent definition that an e-process is a process upper bounded by a family of test martingales.)

Intermezzo: Why are filtrations important?

Pérez-Ortiz, Lardy, De Heide, Grünwald - AoS '24

- Suppose data $X_1, X_2, \ldots$ is sample from $\mathcal{N}(\mu, \sigma^2)$, and let $\delta = \mu/\sigma$. Consider testing $\mathcal{H}_0 : \delta = \delta_0$ vs. $\mathcal{H}_1 : \delta = \delta_1$

- Let $\mathbb{F}$ be the `full' data filtration, and let $\mathbb{G}$ denote the scale-invariant **coarsening** of $\mathbb{F}$:
$$\mathcal{G}_t = \sigma\left(\frac{X_1}{|X_1|}, \ldots, \frac{X_t}{|X_1|}\right), \forall t$$

- In $\mathbb{G}$, an (in a certain sense optimal) e-process $(\mathfrak{e}_t)_{t\geq 0}$ for this problem can be derived.

- This process is not an e-process w.r.t. $\mathbb{F}$:

  If $\tau^{\mathbb{F}} = 1 + \mathbf{1}\{|X_1| \in [0.44,1.7]),$ then $\mathbb{E}[\mathfrak{e}_{\tau^{\mathbb{F}}}] \approx 1.19 > 1.$

The e-process: any-time valid inference (4)

- **Ville's inequality**

  If E is an e-process for $\mathscr{P}$, then

  $P\left(\exists t \in \mathbb{N} : E_t \geq \frac{1}{\alpha}\right) \leq \alpha$ for every $P \in \mathscr{P}$ and $\alpha \in [0,1]$

  $\Leftrightarrow \sup_{P \in \mathscr{P}} \left(\sup_{t \in \mathbb{N}} E_t \geq \frac{1}{\alpha} \right) \leq \alpha$ for every $\alpha \in [0,1]$.

Multiple testing

Bringing closure to FDR control: a general principle for multiple testing

Ziyu Xu, Aldo Solari, Lasse Fischer, Rianne de Heide, Aaditya Ramdas and Jelle Goeman

https://arxiv.org/pdf/2509.02517

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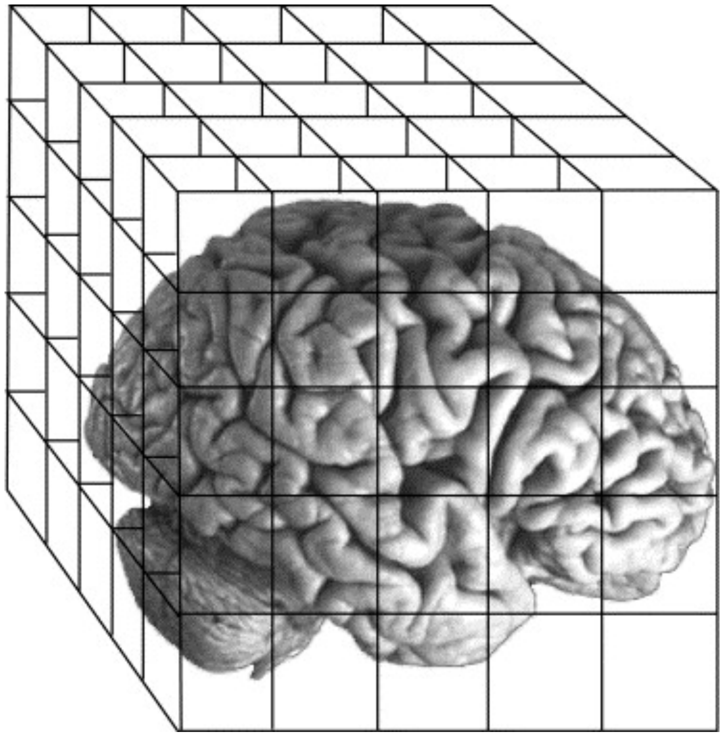

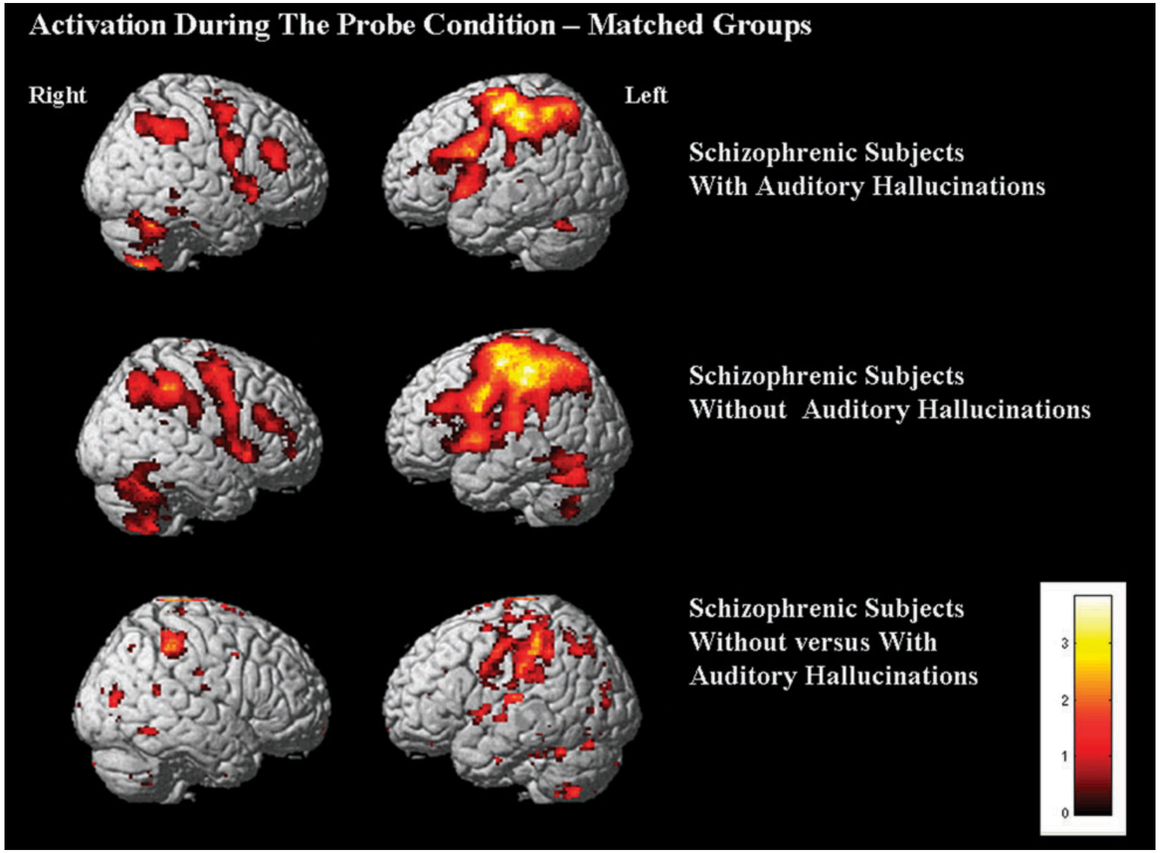

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Multiple testing: the problem

- If we test n true null hypotheses at level α, then on average we will (falsely) reject αn of them.
- Examples:
  - testing whether some of 20.000 genes are linked to a disease
  - fMRI: 100.000 voxels
  - DNA methylation: 500.000 sites
- We need other measures of acceptance/rejection errors.
- We need statistical procedures to control these measures of errors.

Error rates

$N \subseteq [m]$ hypotheses are true null; the rest are potential discoveries

Most famous error rates:

- Familywise error rate (FWER): $P(|R \cap N| > 0)$
- False Discovery rate (FDR): $\mathbb{E}\left(\frac{|R \cap N|}{R}\right)$

General form

Control some expected loss: $\mathbb{E}(f_N(R))$

FWER history

- Closure principle: necessary and sufficient for the construction of valid methods (Sonnemann, 1982, 2008)
- Any method that controls FWER is a special case of a closed testing procedure
- Challenged by the Partitioning Principle (Finner and Strassburger, 2002)
- They are equivalent: Goeman et al. (2021)

- Summary: well-established theory; closure principle is necessary and sufficient

False Discovery Proportion (FDP) history

- Genovese and Wasserman (2006) and Goeman and Solari (2011) have extended closed testing to control of false discovery proportions (FDPs)

- Goeman et al. (2021) showed that all methods controlling a quantile of the distribution of FDP are either equivalent to a closed testing procedure or are dominated by one, extending the Closure Principle to all methods controlling FDP.

- Summary: also for FDP the Closure Principle is key.

The closure principle

- Drawing

Why is the closure principle nice?

- reduces the complex task of constructing a multiple testing method to the simpler task of choosing hypothesis tests for intersection hypotheses
- helps to handle complex situations such as restricted combinations e.g. $H_1 : \mu_1 = \mu_2; H_2 : \mu_2 = \mu_3; H_3 : \mu_1 = \mu_3$
- methods constructed using closed testing often allow for some user flexibility, permitting researchers to modify some aspects of the multiple testing procedure post hoc without compromising error control

FDR history

- Blanchard and Roquain (2008) formulated two quite general sufficient conditions, self-consistency and dependence control, under which, if both hold, FDR control is guaranteed.

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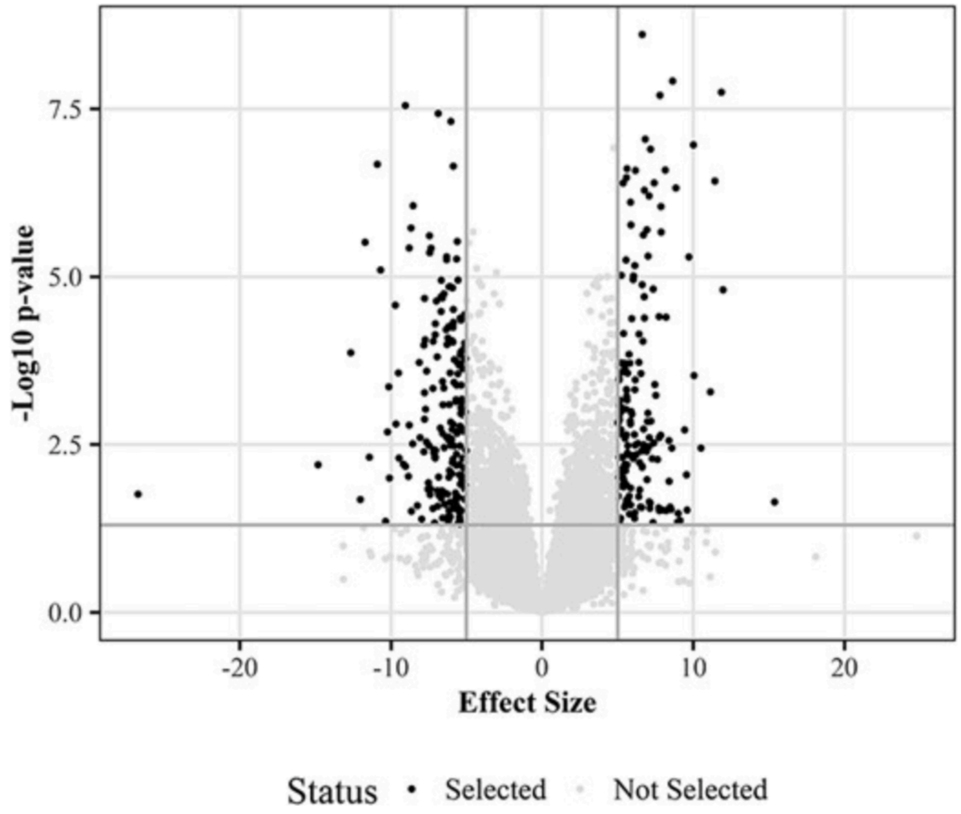
- Blanchard and Roquain (2008) formulated two quite general sufficient conditions, self-consistency and dependence control, under which, if both hold, FDR control is guaranteed.
- However, self-consistency is sufficient but not necessary for FDR control: Solari and Goeman (2017) show uniform improvements of self-consistent methods by a non-self-consistent method.
- Summary: no necessary & sufficient principles. Ad-hoc methods. No general theory. No flexibility.
- No post-hoc user flexibility. Why is that problematic? Example on the next slide.

FDR control and volcano plots: no guarantees!

Ebrahimpoor & Goeman (2021)



The e-closure principle

How to design a multiple testing procedure

- e-Closure
  A general recipe for making multiple testing methods
- Building blocks
  Intersection hypotheses and e-values
- Contributions

  - Recovers the Closure Principle for FWER
  - Extends to FDR
  - Uniformly improves a.o. eBH and BY
  - Introduces unprecedented flexibility in multiple testing

Intersection hypotheses

Intersection hypothesis

For $S \subseteq [m], H_S = \bigcap_{i \in S} H_i$, which is true iff all $H_i, i \in S$ true

The e-collection

$E = (e_S)_{S \subseteq [m]}$: local e-values such that $\mathrm{E}(e_N) \leq 1$

Sufficient

Each e_S is an e-value for $H_S, S \subseteq [m]$

The e-Closure Principle

- The e-Closed Procedure

$$\mathscr{R}_\alpha(E) = \left\{ R \subseteq [m] : \alpha e_S \geq \mathrm{f}_S(R) \quad \forall S \subseteq [m] \right\}$$

- The e-Closure Principle

R controls $\mathrm{E}(\mathrm{f}_N(R)) \leq \alpha$ iff $R \in \mathscr{R}_\alpha(E)$ for e-collection E

- Simultaneous control

$\mathrm{E}(\mathrm{f}_N(R)) \leq \alpha$ simultaneously over $R \in \mathscr{R}_\alpha(E)$: $\quad \mathrm{E}\left(\max_{R \in \mathscr{R}_\alpha(E)} \mathrm{f}_N(R) \right) \leq \alpha$

Post hoc error rate

- All error rates

$$\mathscr{F} = \{\text{all functions } \mathrm{f}_N(R)\}$$

- Simultaneous (= post hoc) choice of error

$$\mathrm{E}\left(\sup_{\mathrm{f}\in\mathscr{F}} \max_{R\in\mathscr{R}^{\mathrm{f}}_{\alpha}(E)} \mathrm{f}_N(R)\right) \leq \alpha$$

- So: possible to switch from FWER to FDR if not much signal present

BY vs closed BY in standard data sets

Dataset	m	BY / $\overline{BY}$ rejections $\alpha = 5\%$	$\alpha = 10\%$	source
APSAC	15	3 / 3	3 / 5	BH '95
NAEP	34	6 / 8	8 / 11	BH '00
PADJUST	50	12 / 15	17 / 20	`p.adjust`
PVALUES	4289	129 / 145	225 / 275	`fdrtool`
VANDEVIJVER	4919	614 / 677	779 / 866	Goeman Solari '14
GOLUB	7128	617 / 648	743 / 799	Efron Hastie '16

More properties: post hoc α

- Choose rejected set post hoc
- Choose error loss post hoc
- One step further: choose α post hoc (Koning 2023)
- Requires: e-value does not depend on α. Then we have:

$$\mathrm{E}\left(\sup_{\alpha \in (0,1)} \sup_{\mathrm{f} \in \mathscr{F}} \max_{R \in \mathscr{R}_\alpha^{\mathrm{f}}(E)} \frac{\mathrm{f}_N(R)}{\alpha} \right) \leq 1$$

More properties: restricted combinations

- Logically related hypotheses, for example pairwise combinations

$$H_{1=2} : \mu_1 = \mu_2; \quad H_{1=3} : \mu_1 = \mu_3; \quad H_{2=3} : \mu_2 = \mu_3$$

- Logical relationships = gain in power

  Up to now only known for FWER, unknown for FDR

Summary: e-Closure

- General Necessary and Sufficient Principle: unites all multiple testing methods
- Simplifies multiple testing: Choose how to summarise evidence against H_S; rest is computation
- Flexibility: Simultaneous over rejected sets, error rates, α
- Power: Uniformly improves known methods

A general recipe for making multiple testing methods

A problem

Carefree multiple testing with e-processes.

Yury Tavyrikov, Jelle J. Goeman, and Rianne de Heide.

https://arxiv.org/abs/2501.19360

Testing a single hypothesis: no problem

- One e-process: $E_1, E_2, \ldots$
- Suppose we already gathered E_{t+1} before properly evaluating E_t.
- If $E_t \geq 1/\alpha$ but $E_{t+1} < 1/\alpha$, we can still go back to E_t and reject H_0, because Ville's inequality states

$$\sup_{P \in \mathscr{P}} \left(\sup_{t \in \mathbb{N}} E_t \geq \frac{1}{\alpha} \right) \leq \alpha$$

Testing multiple hypotheses: a problem (1)

- Let's focus on e-BH for FDR control.

- Consider K parallel e-processes $E_t^1, \ldots, E_t^K$

- e-BH guarantees: $\sup_t \text{FDR}(E_t^1, \ldots, E_t^K) \leq \alpha$

Testing multiple hypotheses: a problem (2)

- Consider $K = 2$, then e-BH rejects any hypothesis for which the e-value exceeds $2/\alpha$, and both hypotheses if both e-values exceed $1/\alpha$.

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- Gathering more data for both e-processes, we could arrive at

  $E_{t+1}^1 < 1/\alpha$ and $1/\alpha \leq E_{t+1}^2 < 2/\alpha$.

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- Gathering more data for both e-processes, we could arrive at

  $E_{t+1}^1 < 1/\alpha$ and $1/\alpha \leq E_{t+1}^2 < 2/\alpha$.

- Now, if we would not have gathered one more data point for E^1, then we would have rejected both hypotheses.

FDR control for e-processes

- e-BH guarantees: $\sup_t \mathrm{FDR}\left(E_t^1, \ldots, E_t^K\right) \leq \alpha$
- But we would like to control:

  $$\mathrm{FDR}\left(\sup_t E_t^1, \ldots, \sup_t E_t^K\right) \leq \alpha$$

- We would like to argue that this is the proper FDR criterion for e-processes.
- *(Note: in this example I focus on FDR with e-BH, but we can show the same for other methods, including e-closed methods and FWER.)*

A proposition

Working with the running maxima of e-processes

- Clearly, if we work with $M_t^k := \max_{1 \le s \le t} E_s^k$, the running maximum processes, we obtain a non-decreasing sequence of sets of rejected hypotheses.

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- In case the e-processes are independent, $1/M_t^k$ is a valid p-value (p-process), thus, we obtain FDR control via classical BH.

Proposition

(Tavyrikov, Goeman, De Heide, ’25)

- **Proposition**

  The cumulative maximum e-BH procedure applied to arbitrarily dependent e-processes does not control the FDR at level α.

A solution

Adjusters

- An (admissible) adjuster is an increasing function $A : [1,\infty] \to [1,\infty]$ that is right-continuous, $A(\infty) = \lim_{e\to\infty} A(e) = \infty$ and $\int_1^\infty A(e)/e^2 \mathrm{d}e = 1$.

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- If $M_t^k := \sup_{1 \leq s \leq t} E_s^k$ is the running maximum of an e-process, then

  $(A(M_t^k))_{t \geq 0}$ is also an e-process.
- Since the adjusted process is non-decreasing, it will never result in regretting having collected more evidence.

The adjusted running maximum process has FDR-sup $\leq \pi_0 \alpha$ (Tavyrikov, Goeman, De Heide, '25)

Let $\mathscr{R}^{\text{e-BH}}(\mathbf{e})$ denote the e-BH rejection set based on a set of e-values

$\mathbf{e} = e^1_{t_1}, \ldots, e^K_{t_K}$, which can be obtained from different time points $t_1, \ldots, t_K$.

I.e., $j \in \mathscr{R}^{\text{e-BH}}(\mathbf{e})$ if and only if $e^j_{t_j} > m/(\alpha\,|\,\mathscr{R}^{\text{e-BH}}(\mathbf{e})\,|\,)$. Let A be an adjuster,

so that $A\left(\sup_{t \leq s}(e^j_t)\right)$ is an adjusted running maximum process.

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Tavyrikov, Goeman, De Heide, '25

Then the FDR-sup at time s is

$$\text{FDR-sup}_s = \sum_{j \in H_0} \frac{\mathbf{1}\left\{ A\left(\sup_{t \leq s}(e_t^j)\right) \geq \frac{K}{\alpha |\mathscr{R}^{\text{e-BH}}(\mathbf{e})|} \right\}}{|\mathscr{R}^{\text{e-BH}}(\mathbf{e})| \vee 1}$$

$$\leq \sum_{j \in H_0} \mathbb{E}\left[A\left(\sup_{t \leq s}(e_t^j) \right) \frac{\alpha \frac{|\mathscr{R}^{\text{e-BH}}(\mathbf{e})|}{K}}{|\mathscr{R}^{\text{e-BH}}(\mathbf{e})| \vee 1} \right] \leq \frac{\alpha}{K} \sum_{j \in H_0} \mathbb{E}\left[A\left(\sup_{t \leq s}(e_t^j) \right) \right] \leq \frac{K_0}{K} \alpha \, .$$

Do we *need* adjusters?

- **Theorem (informal)**: The function is *necessarily* an adjuster, as long as it is an *increasing* function that maps the running maximum process to an e-process.

Interesting next problem

Forthcoming work by Tavyrikov, Goeman, De Heide

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- But under the common filtration $\mathscr{G}_t = \sigma(\mathbf{1}\{U \leq 1/t\}, \mathbf{1}\{U \leq 1/(2t)\})$, they are not jointly valid: with $\tau = \dfrac{1}{U}, \quad E_\tau^{(1)} = \dfrac{1}{U},$ we get $\mathbb{E}[E_\tau^{(1)}] = \infty$.
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- So: individually valid, but not jointly valid under a richer common filtration.
- This happens almost all the time, and for more and more realistic e-processes, finding a common *and* useful filtration turns out to be very hard or impossible.

Conclusion

- We found a unifying necessary and sufficient principle for multiple testing:

  e-closure

- We argue that any-time valid multiple testing procedures should be carefree, meaning that the researcher should never lose out on any rejections because they chose to gather more data for one or more e-processes.

- Carefree multiple testing procedures can operate on the running maxima of e-processes. They can be turned into e-processes by using adjusters.

- Forthcoming work with Yury Tavyrikov and Jelle Goeman: general theory including tight tail bounds on monotone non-decreasing e-processes, allowing for efficient multiple testing with adjusted maxes of e-processes, that can be used together with e-closure.

- New problem: finding common filtrations for several e-processes such that all are valid; and also practicable.